

Global Automation

Humanoid Robotics: The technology frontier in 2026



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Humanoid robotics is entering a phase of accelerated development and at the threshold of large-scale commercial adoption. This note explains the current frontier of the industry and provides benchmarks for investors to assess progress and the numerous claims made by industry players.

We highlight the frontier and future directions in five dimensions (Exhibit 1), along which, examples of progresses from notable players are shown in Exhibit 2. In **locomotion**, stable walking on flat surface, which was impressive two years ago, has progressed to motions that require high dynamic full-body control and adaptive interaction with environment. **Manipulation tasks** have advanced from single-step pick-and-place to increasing dexterity, longer task sequence, and industrial level reliability, although in all these areas, development remains in the early stage. In **the level of autonomy**, remote control is largely abandoned, and autonomous short-horizon tasks and pre-defined long-horizon tasks are currently the most common. Long-horizon autonomous planning, skill transfer, and task generalization are just emerging and the focus of future development. The **robotic brain** paradigm has evolved from LLM/VLM to vision-language-action (VLA) models (2024-2025), which are enhanced by World Action Models (WAM) from 2026 onward. **Data modality** has greatly diversified, and the general trend is to move away from teleoperation (high cost, low efficiency, and hardware dependent) and prioritize egocentric, platform-agnostic data (such as human videos), and the complementary use of non-vision data.

At the center of these progresses is the evolution of robotic brain models to use World Model as the “planner”. In essence, VLA is “language to action”, while WAM is “action based on environmental state, context, and experience”. WAM predicts physically feasible future states (Exhibit 6) and then generates future actions based on those predictions. It is analogous to human brains imagining outcomes before taking actions (Exhibit 4). This way, WAM promises to improve task generalization and performance robustness (Exhibit 8, Exhibit 10, Exhibit 11). Humanoid robot players are quickly embracing the idea of teaching robots to “imagine the future”, by either developing standalone WAM or embedding World Model within an existing VLA (Exhibit 9). Currently, challenges include slower inference speed compared to VLA and the very limited non-vision data such as tactile and material properties to train WAMs.

The rise of WAM may reshape the industry ecosystem and competitive dynamics. WAM emphasizes data diversity with regard to both robot actions and environment. This type of data is more likely to be available from open sources and agnostic of hardware platforms. This potentially lowers the burden of post-training data collected from teleoperation or deployed robots. Therefore, the competitive moat partially shifts from “data ownership” to superior model architecture and data consolidation capabilities. This shift conceivably benefits independent brain / foundational WAM developers such as Physical Intelligence (private) and Nvidia. They will likely complement the brain-body integrated robot OEMs such as Figure AI, Agibot, and LimX (all private). We imagine that robot brain will evolve to be analogous to EV battery and power train — some best OEMs choose to have their own, and the rest find it better to purchase from the best third-party suppliers. The WAM paradigm may also favor players that fill the critical physical data gaps for World Models, such as PaXini (private) in tactile sensing.

DETAILS

EXHIBIT 1: The frontier and future directions in five dimensions of humanoid robots

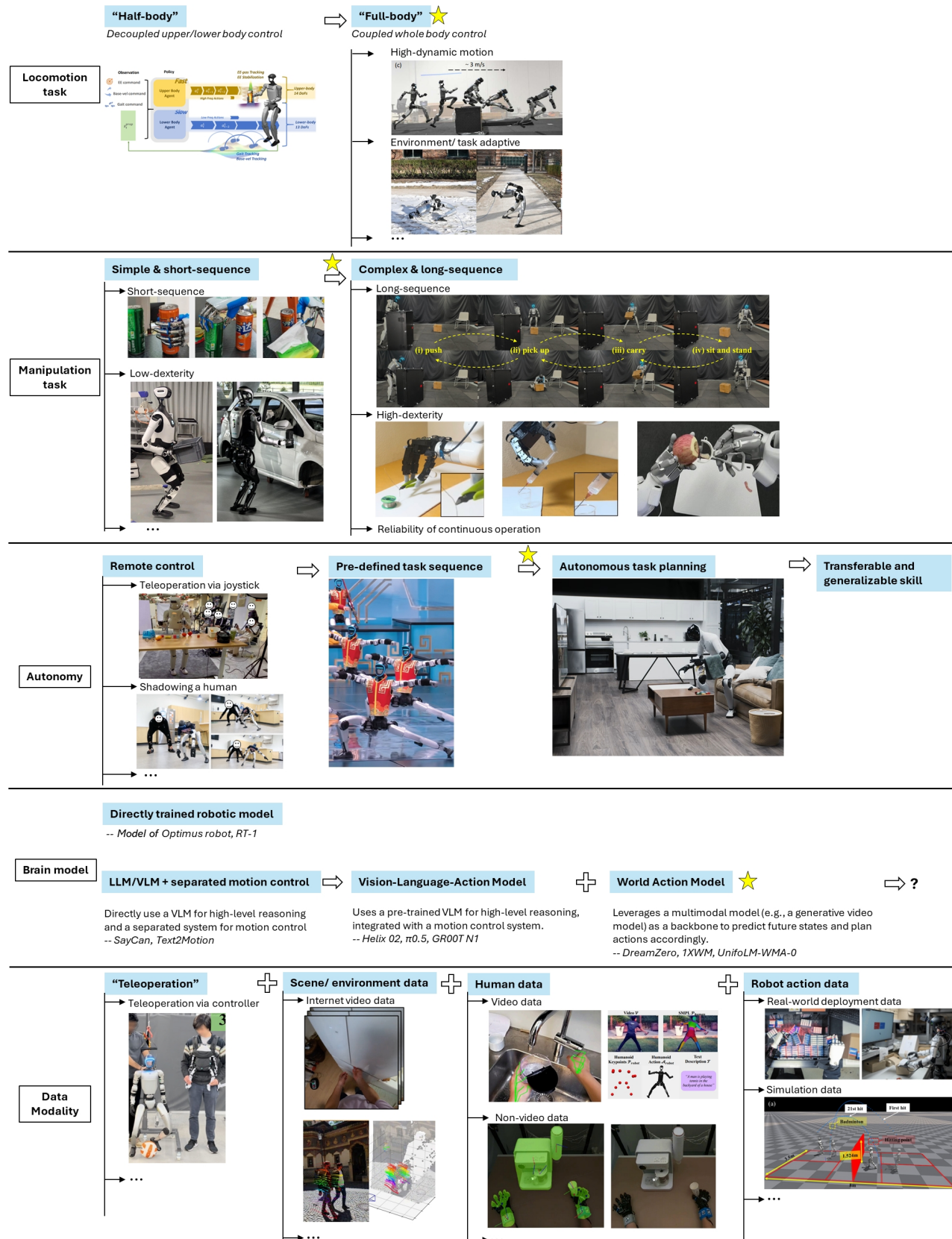
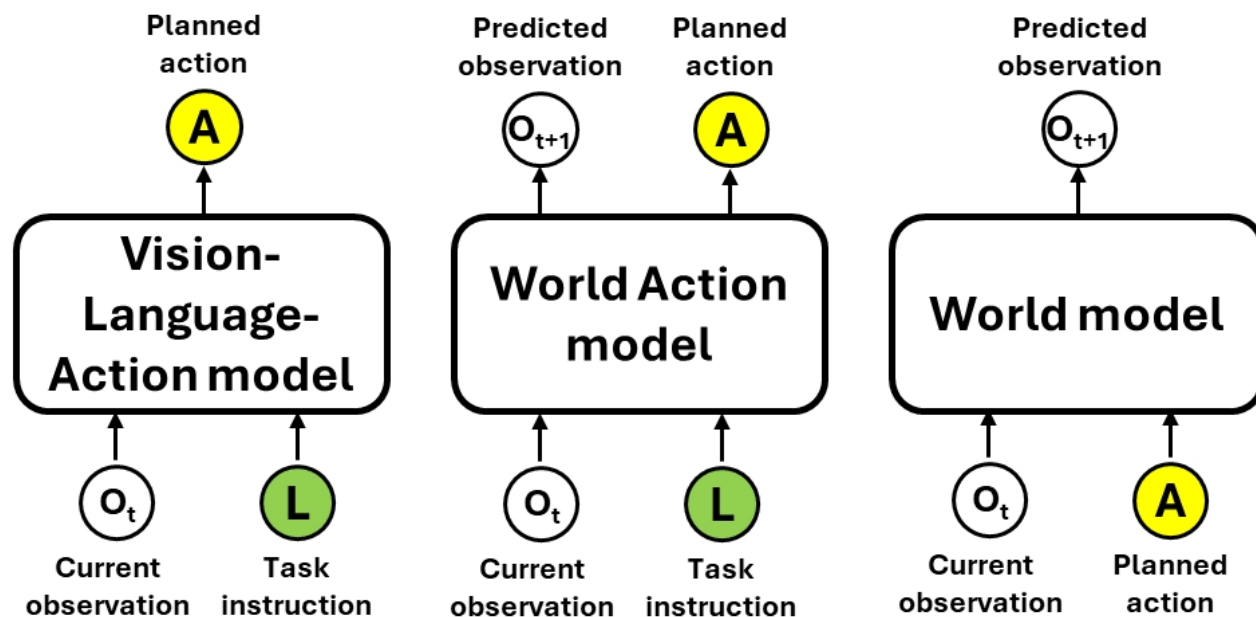


EXHIBIT 2: **Examples of latest demonstration in the key frontiers from notable humanoid players**

Company	Locomotion		Manipulation		Brain & world model
	High dynamic	Adaptive	Dexterity & long-horizon	Reliability	
Unitree	Spring Festival Gala Performance ¹	Chase wild boars across varied terrain ²			UnifoLM-WMA-0 model
Agibot				Continuous inspection-line operation ³	
LimX			Fully autonomous housekeeping ⁴		COSA 0.5 model
Paxini					Integrates Tactile Simulator into Nvidia Isaac Sim
Figure AI			Fully autonomous housekeeping ⁵	Continuous package sorting ⁶	Helix 02 model
Boston Dynamics	"The Ghost Rabona" — a deceptive football trick shot ⁷		Whole-fridge transport ⁸		
Physical Intelligence					$\pi_{0.7}$ model
Nvidia					DreamZero model

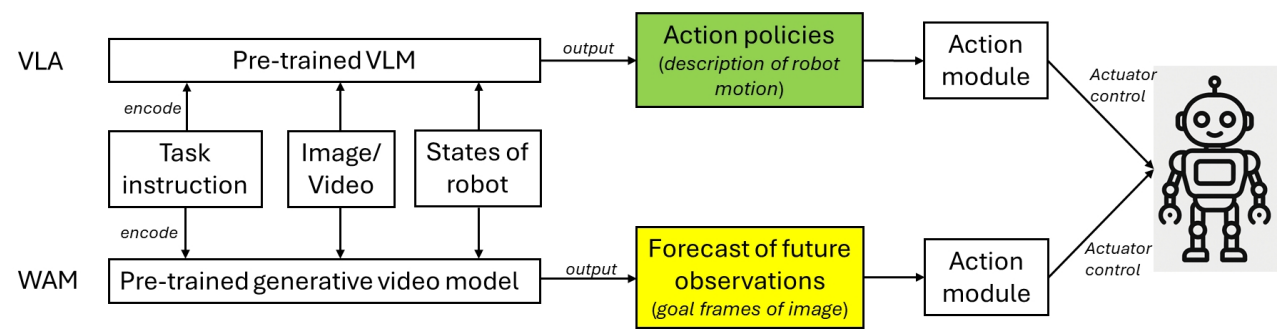
Note: Please see the link of each example by order in the last page of this note. Unitree, Agibot, Paxini, Figure AI, and Physical Intelligence are private. Boston Dynamic is a subsidiary of Hyundai Motor Group which is not covered by Bernstein. Nvidia is covered by Bernstein U.S. semiconductors team.

Source: Bernstein analysis

EXHIBIT 3: **Compared to a VLA, world models forecast the outcome of evolution of environment ("predicted observation"), which is used to generate future actions.**

Source: Bernstein analysis

EXHIBIT 4: **WAM is analogous to human brains imagining outcomes before taking actions, while VLA outputs the next action directly.**



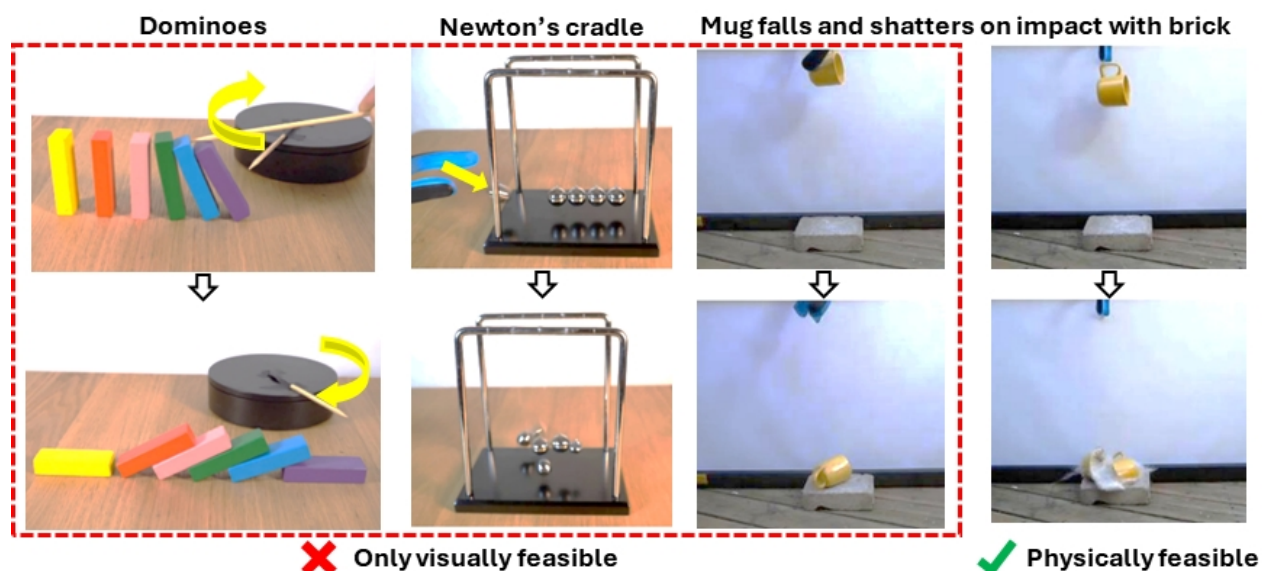
Source: Bernstein analysis

EXHIBIT 5: **Three main types of world models and their applications**

World model	“Render”	“Simulator”	“Planner”
Rendering visual scene	++	+	+
Predicting future states	+	++	++
Reasoning and action planning			++
Models	SORA, Kling, Veo, Wan-Video	Cosmos	DreamZero, 1XWM, UnifoLM-WMA-0

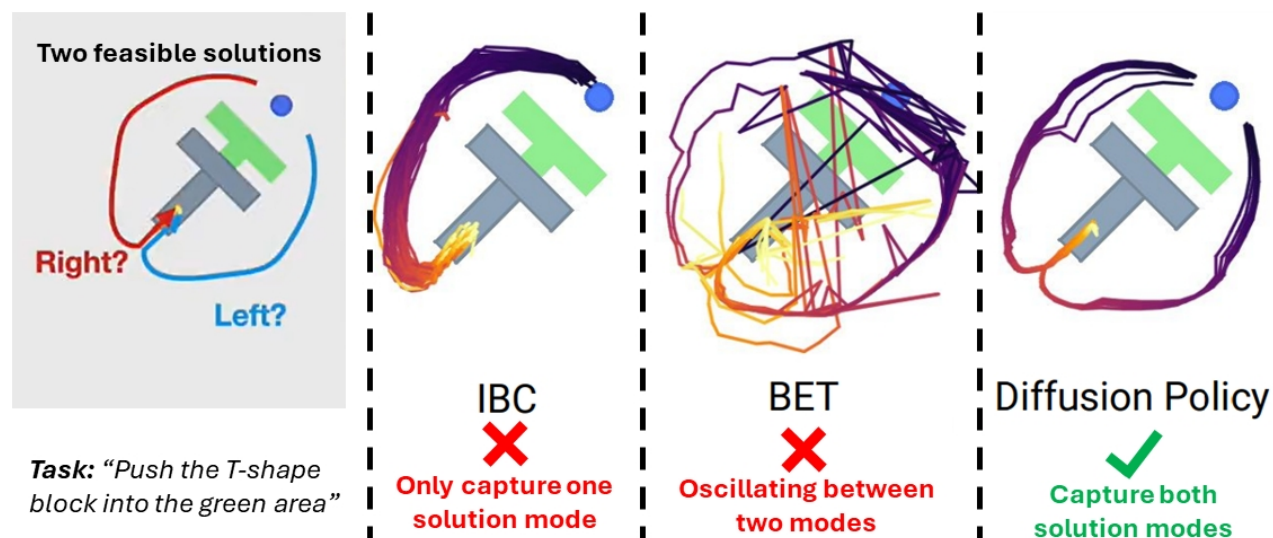
Source: Bernstein analysis

EXHIBIT 6: “Render” world models generate visually plausible outputs, while “simulator” and “planner” world models embed learned physical laws to produce physically feasible outputs.



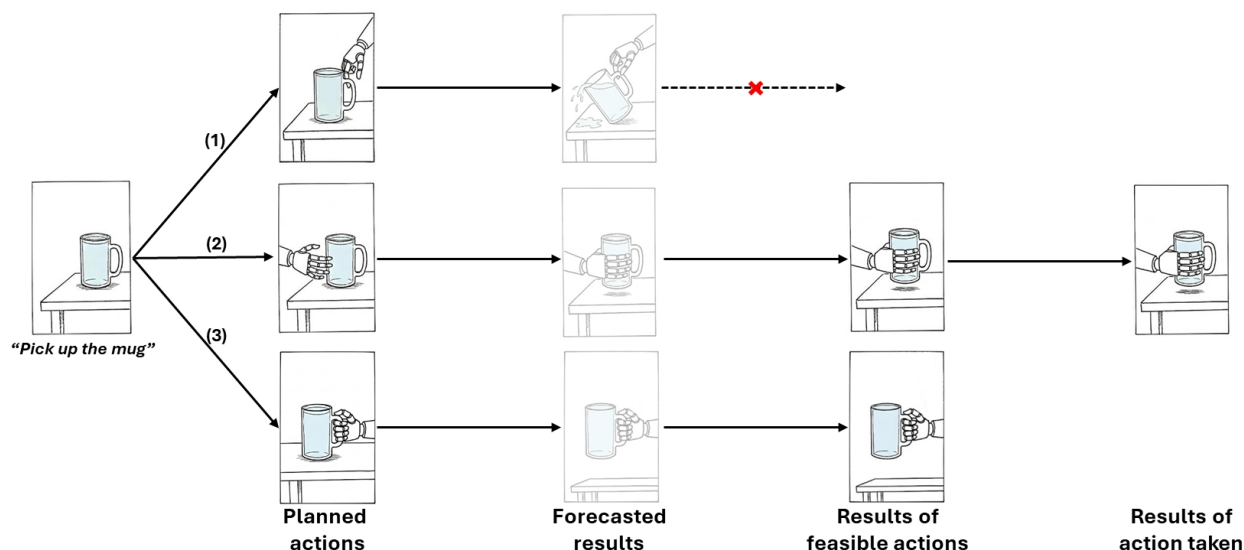
Source: “PhyGround: Benchmarking Physical Reasoning in Generative World Models” by Juyi Lin and etc., “PhyWorld: Physics-Faithful World Model for Video Generation” by Pu Zhao and etc., Bernstein analysis.

EXHIBIT 7: Diffusion policy models can capture multiple solution modes and select one as the final one, while other model types may collapse toward a single mode or oscillate between modes, limiting task generalization capability.



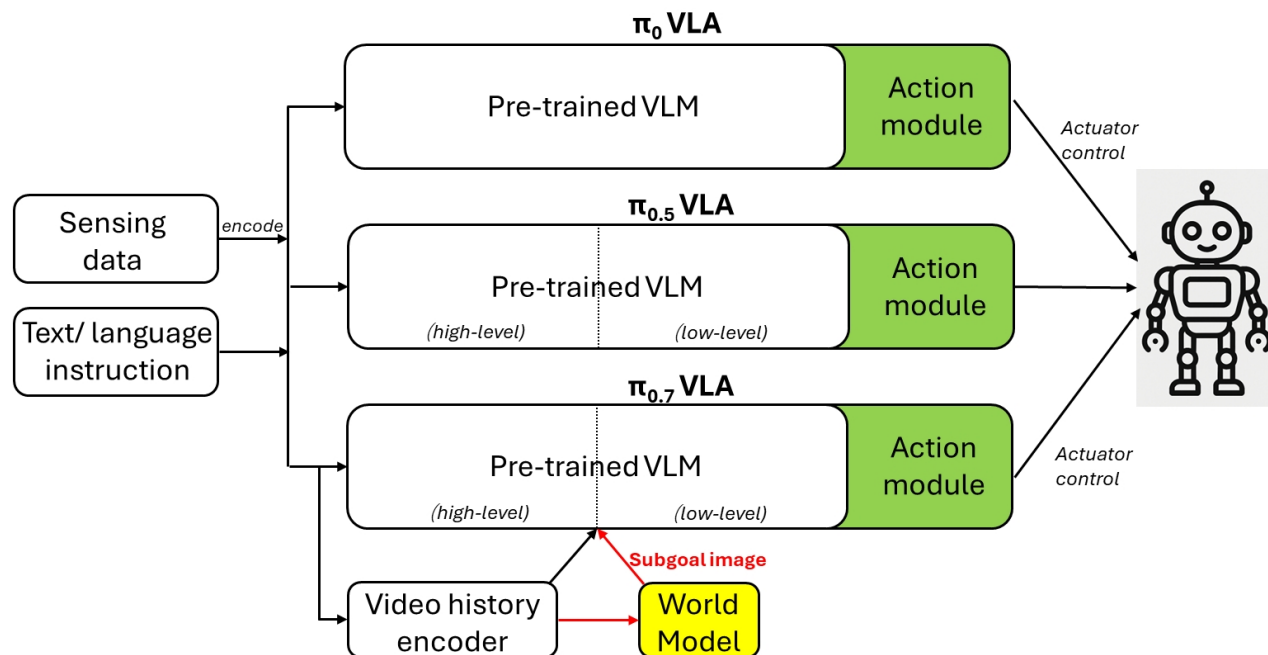
Source: “Diffusion Policy: Visuomotor Policy Learning via Action Diffusion” by Cheng and etc., Bernstein analysis

EXHIBIT 8: **World Action Models with a diffusion-policy backbone can output physically feasible policies and better handle tasks with multiple valid solutions – such as picking up a mug filled with water.**



Source: Bernstein analysis

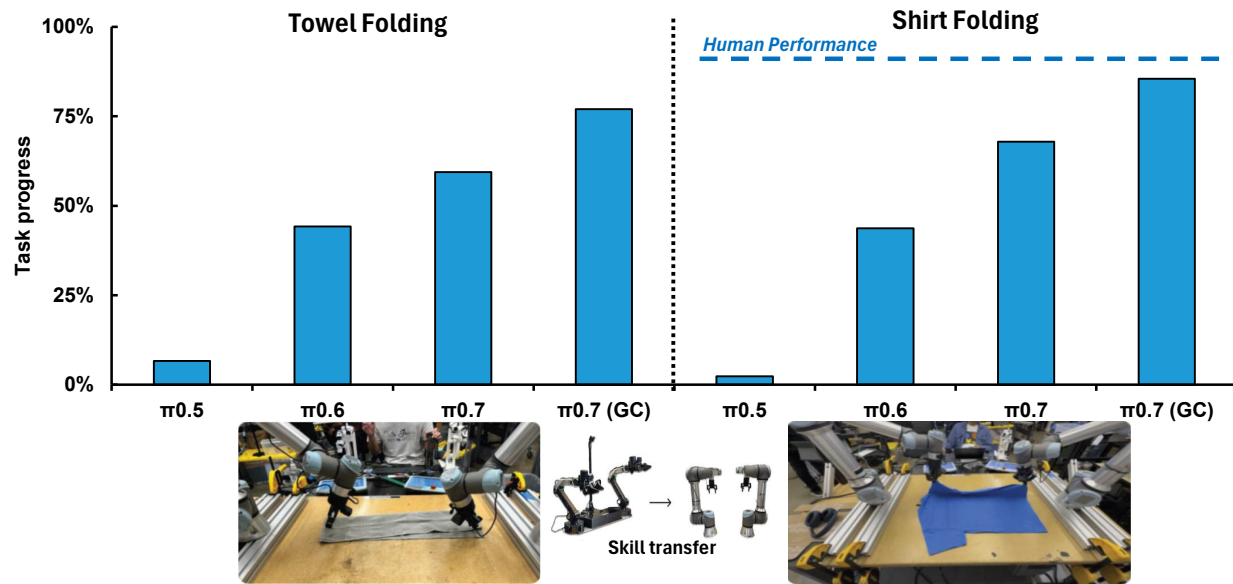
EXHIBIT 9: **Evolution of Physical Intelligence's VLA models over time**



Note: Physical Intelligence is private.

Source: Physical Intelligence's website and reports, Bernstein analysis

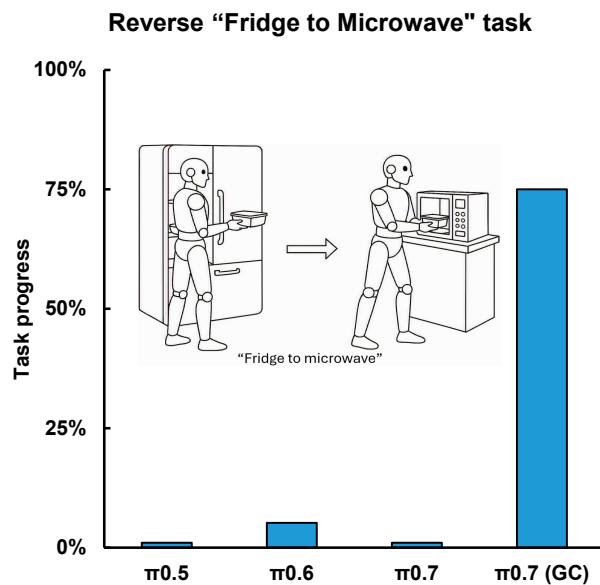
EXHIBIT 10: The adoption of the world model not only improved $\pi 0.7$'s performance in complex tasks but also enhanced its capability for skill transfer between robots.



Note: Performance of using subgoal image generation with world model is labeled as $\pi 0.7$ (GC) in the plot. Physical Intelligence is private. Please refer to the following note for a detailed definition of task progress by Physical Intelligence: <https://arxiv.org/pdf/2504.16054>.

Source: Physical Intelligence, Bernstein analysis

EXHIBIT 11: With the embedded World Model, $\pi 0.7$ (GC) breaks strong training dataset biases – trained on “microwave to fridge” and tested on “fridge to microwave”



Note: Performance of using subgoal image generation with world model is labeled as $\pi 0.7$ (GC) in the plot. Physical Intelligence is private. Please refer to the following note for a detailed definition of task progress by Physical Intelligence: <https://arxiv.org/pdf/2504.16054>.

Source: Physical Intelligence, Bernstein analysis

Academic paper as the source for Exhibit 1, as follows: “Hold My Beer: Learning Gentle Humanoid Locomotion and End-Effector Stabilization Control” by Yitang Li and etc., “Perceptive Humanoid Parkour: Chaining Dynamic Human Skills via Motion Matching” by Zhen Wu and etc., “Learning Getting-Up Policies for Real-World Humanoid Robots” by Xialin He and etc., “RGMP: Recurrent Geometric-prior Multimodal Policy for Generalizable Humanoid Robot Manipulation” by Xuetao Li and etc., “Load-Aware Locomotion Control for Humanoid Robots in Industrial Transportation Tasks” by Lequn Fu and etc., “Path and Motion Optimization for Efficient Multi-Location Inspection with Humanoid Robots” by Jiayang Wu and etc., “LESSMIMIC: Long-Horizon Humanoid Interaction with Unified Distance Field Representations” by Yutang Lin and etc., “MANA: Dexterous Manipulation of Articulated Tools” by Zhaocheng Yin and etc., “Towards Human-Like Manipulation through RL-Augmented Teleoperation and Mixture-of-Dexterous-Experts VLA” by Tutian Tang and etc., “TWIST: Teleoperated Whole-Body Imitation System” by Yannjie Ze and etc., “CHILD (Controller for Humanoid Imitation and Live Demonstration): a Whole-Body Humanoid Teleoperation System” by Noboru Myers and etc., “EgoVLA: Learning Vision-Language-Action Models from Egocentric Human Videos” by Ruihan Yang and etc., “Learning from Massive Human Videos for Universal Humanoid Pose Control” by Jiageng Mao and etc., “Cortex 2.0: Grounding World Models in Real-World Industrial Deployment” by Adriana Aida and etc., “Humanoid Whole-Body Badminton via Multi-Stage Reinforcement Learning” by Chenhao Liu and etc., “RoboPaint: From Human Demonstration to Any Robot and Any View” by Jiacheng Fan and etc., “Structured World Models from Human Videos” by Russell Mendonca and etc., “Stereo4D: Learning How Things Move in 3D from Internet Stereo Videos” by Linyi Jin and etc..

Link to each example in Exhibit 2:

1. <https://thekidshouldseethis.com/post/martial-arts-robots-viral-video-2026-spring-festival-gala> ;
2. <https://apnews.com/video/humanoid-robot-chases-wild-boars-in-the-polish-capital-warsaw-189795eb9cc641f7a5b5d071632e7c01>
3. <https://interestingengineering.com/videos/chinas-agibot-is-livestreaming-humanoid-robots-from-a-real-factory>
4. <https://www.limxdynamics.com/zh/news/BK000065>
5. <https://www.figure.ai/>
6. <https://www.rockingrobots.com/figure-ai-claims-200-hour-autonomous-package-sorting-run-with-figure-03/>
7. <https://www.youtube.com/watch?v=V1Lxp-Q6Y9g>
8. <https://www.youtube.com/watch?v=3aQWvdCac9o>

INVESTMENT IMPLICATIONS

Reiterate Outperform for FANUC, Inovance, Harmonic Drive, Cognex, and Keyence; Market-Perform for Estun.

BERNSTEIN TICKER TABLE

Ticker	Rating	16 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
002747.CH (Estun)	M	CNY	42.43	26.00	74.4%	CNY	0.05	0.31	0.32	848.6	138.0	131.2
2715.HK (Estun)	M	HKD	22.70	17.26	NA	CNY	0.05	0.31	0.32	454.0	73.8	70.2
6954.JP (Fanuc)	O	JPY	7,020.00	7,000.00	44.9%	JPY	178.47	207.32	193.15	39.3	33.9	36.3
6324.JP (HDSI)	O	JPY	7,200.00	7,800.00	130.3%	JPY	16.99	57.37	79.51	423.8	125.5	90.6
6861.JP (Keyence)	O	JPY	74,050	86,000	(7.9)%	JPY	1,835.63	2,248.48	2,494.78	40.3	32.9	29.7
300124.CH (Inovance)	O	CNY	60.31	82.00	(34.8)%	CNY	1.87	2.19	2.65	32.3	27.5	22.8
CGNX (Cognex)	O	USD	63.89	75.00	73.3%	USD	0.67	1.46	1.62	95.4	43.8	39.5
ASIAx			1,865.18									
JPL			2,545.21									
SPX			7,457.69									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

6954.JP, 6861.JP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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VALUATION METHODOLOGY**Estun Automation Co Ltd**

We use EV/EBITDA multiple as the primary valuation method. Our price target of RMB26.0 (A-share) and HKD 17.3 (H-share) are based on an EV/EBITDA multiple of 35.9x against our 1-year forward EBITDA estimate of RMB677.2mn. We set the multiple referencing previous cycles but adjust for secular or competitive trends that we believe are moving multiples higher or lower across multiple cycles. We set Estun's H-share TP based on the average A/H share premium. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent target price may deviate from the DCF-implied value.

FANUC Corp

We use EV/EBITDA multiple as the primary valuation method. We set a JPY7,000 target price using an EV/EBITDA multiple of 22.5x against our 1-year forward-looking EBITDA estimates (from the PT date) of JPY 251,396 million. We set the target multiple referencing previous cycles but adjust for secular or competitive trends that we believe are moving multiples higher or lower across multiple cycles. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent target price may deviate from the DCF-implied value.

Harmonic Drive Systems Inc

We use EV/EBITDA multiple as the primary valuation method. We set a JPY 7,800 target price using an EV/EBITDA multiple of 45.5x against our 1-year forward-looking EBITDA estimates (from the PT date) of JPY16,194 mn. We set the target multiple referencing previous cycles but adjust for secular or competitive trends that we believe are moving multiples higher or lower across multiple cycles. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent target price may deviate from the DCF-implied value.

Keyence Corp

We use EV/EBITDA multiple as the primary valuation method. We set a JPY86,000 target price using an EV/EBITDA multiple of 21.5x against our 1-year forward-looking EBITDA estimates (from the PT date) of JPY 831,323 million. We apply our target multiple on the upcoming cycle peak to get the enterprise value, and discount it back to derive our price target. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent target price may deviate from the DCF-implied value.

Shenzhen Inovance Technolo-A

We use EV/EBITDA multiple as the primary valuation method. We set a RMB82 target price using an EV/EBITDA multiple of 24.0x against our 1-year forward-looking EBITDA estimates (from the PT date) of RMB 8971.6 million. We set the multiple referencing previous cycles but adjust for secular or competitive trends that we believe are moving multiples higher or lower across multiple cycles. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent target price may deviate from the DCF-implied value.

Cognex Corp

We use EV/EBITDA multiple as the primary valuation method. We set a USD75.00 price target using an EV/EBITDA multiple of 36.5x against our 1-year forward-looking EBITDA estimates (from the PT date) of USD 333.4 million. We set the multiple referencing previous cycles but adjust for secular or competitive trends that we believe are moving multiples higher or lower across multiple cycles. We use DCF as reference for the company's long-term intrinsic value. As we move along the different stages of a cycle, the time-dependent price target may deviate from the DCF-implied value.

RISKS**Estun Automation Co Ltd**

The risks to our view on Estun are mainly associated with China and global macro economy, including industrial capex cycles, trade frictions, and currency. In addition, the integration of Cloos and realization of planned synergies are important to our thesis and sources of additional risks. The upside risks include 1) faster than expected margin expansion; 2) faster than expected market share gain. The downside risks include 1) weaker than expected automation demands in China; 2) delayed or slower than expected in operating and net margin improvement; 3) delayed or slower than expected market share gain.

FANUC Corp

FANUC's end user markets are cyclical in nature. The risks are mainly associated with the global macro economy and currency. The downside risks include 1) delayed or weaker than expected global automation demand, 2) losing market share to competitors, and 3) appreciation of JPY.

Harmonic Drive Systems Inc

The risks to our view on HDSI are mainly associated with the global macro economy, including industrial capex cycles, competition and currency. As HDSI has >50% of global share in strain wave reducer, potential change in competitive landscape would be a more relevant risk than to other companies. Therefore, the downside risks include: 1) weaker than expected global robot demand, 2) losing market share to competitors, and 3) appreciation of JPY.

Keyence Corp

The risks are mainly associated with the global macro economy. Our analysis shows that company's growth correlates with the overall utilization of manufacturing capacity in the major economies. Fluctuations in utilization may lead to unexpected near term fluctuations in growth rates. A global recession as severe as that of 2009 may even result in decline in Keyence's revenue. In addition, there is currency risk associated with the exchange rate of JPY. Therefore, the downside risks include: 1) weaker than expected overall utilization, 2) delayed or weaker than expected global automation demand, and 3) appreciation of JPY.

Shenzhen Inovance Technolo-A

The risks to our view on Inovance are mainly associated with macro economy, including industrial capex cycles, trade frictions, and currency. The downside risks to our view on Inovance include 1) weaker than expected automation demands in China, 2) weaker or slower than expected share gain in China in segments besides servo motor and VFD, 3) weaker than expected EV demands.

Cognex Corp

Similar to Keyence, the risks are mainly associated with the global macro economy and the overall utilization of manufacturing capacity in the major economies. The downside risks include: 1) delayed or weaker than expected global automation demands, 2) delayed or slower than expected progress in "emerging customer" program, 3) delayed or slower than expected in the integration of Moritex and realization of planned synergies.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price

Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

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- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

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The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

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- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

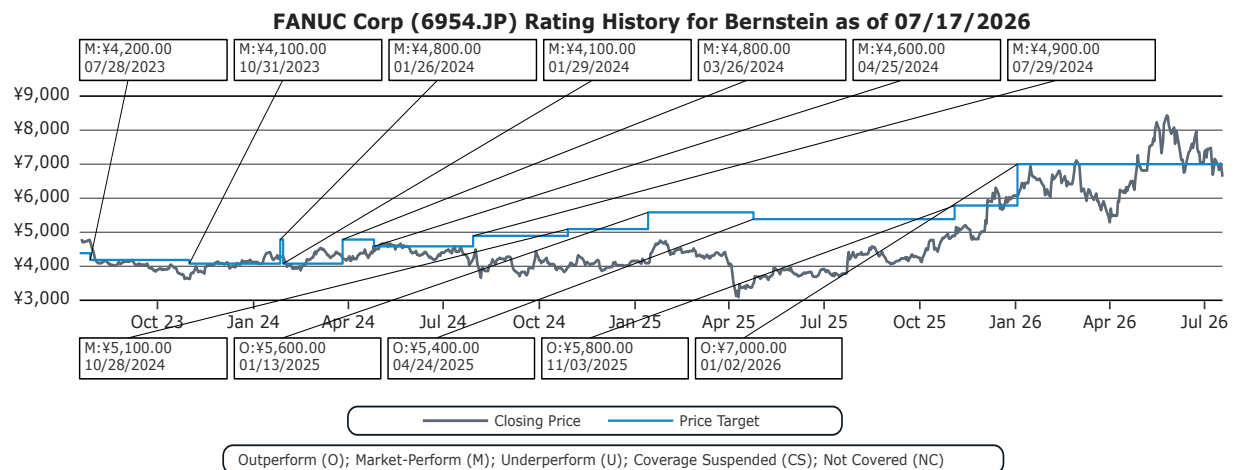
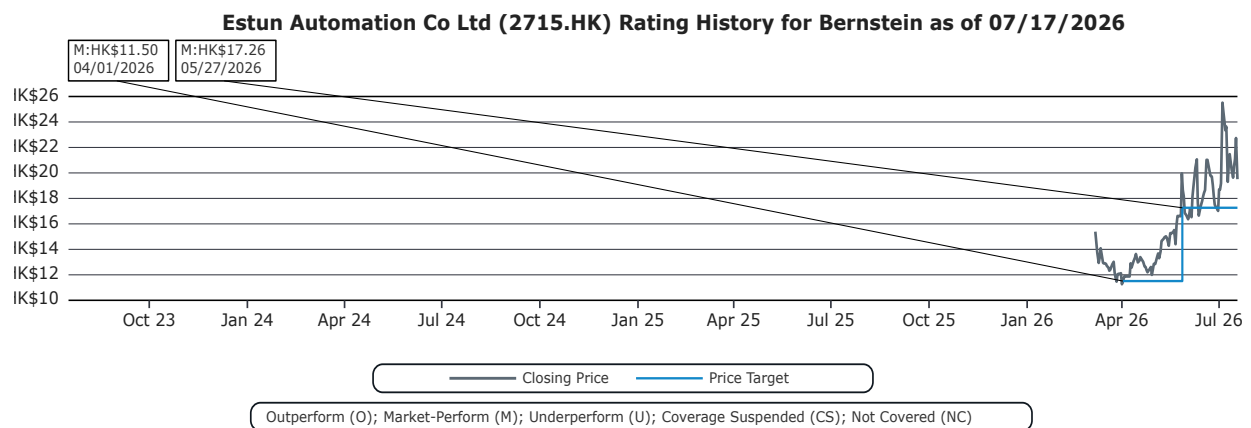
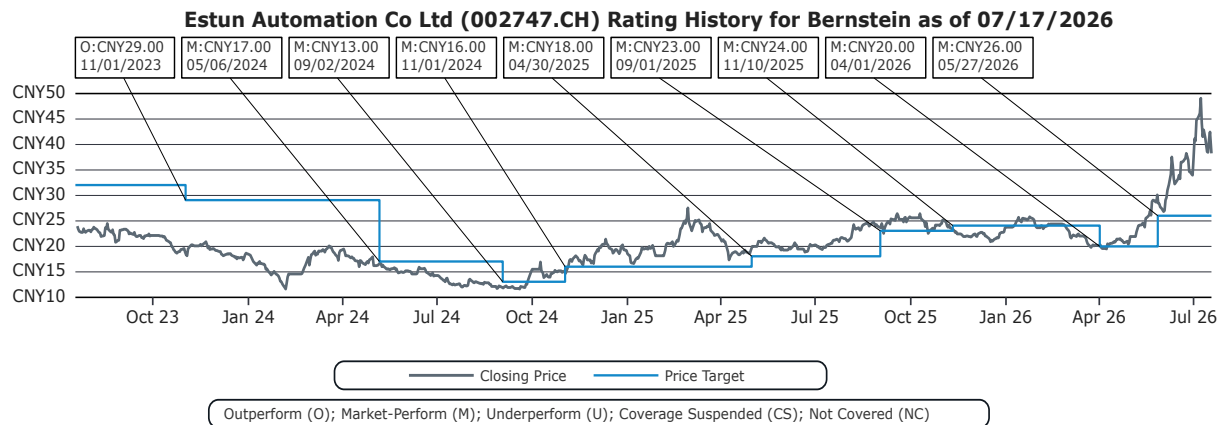
DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

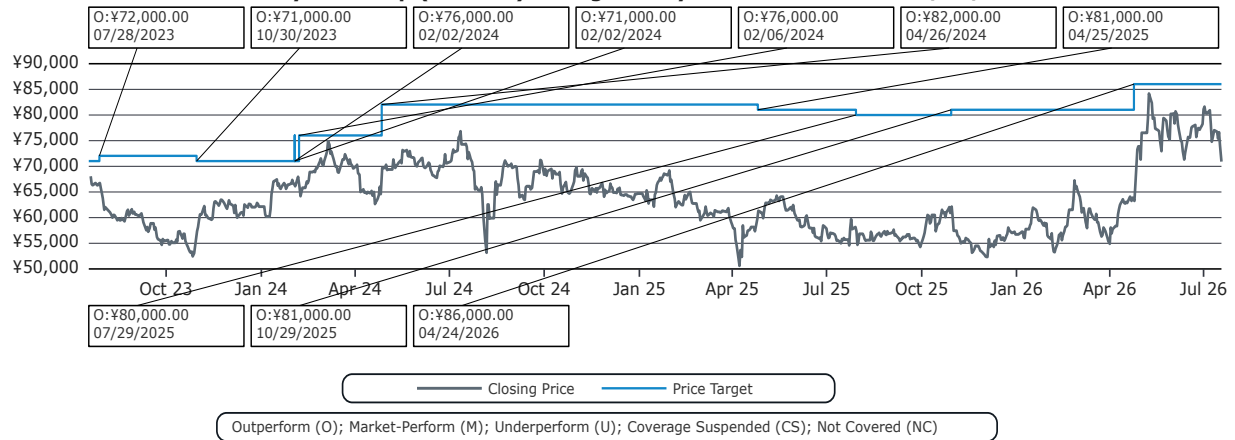
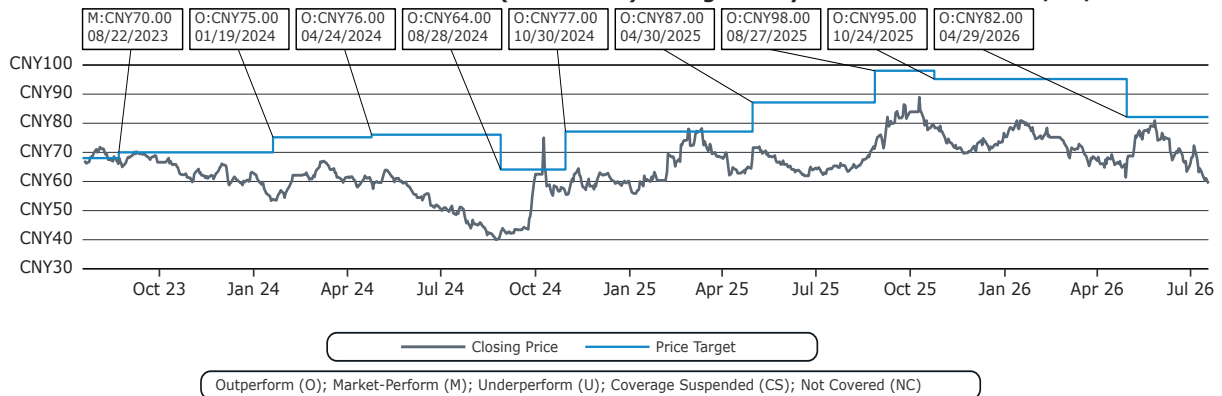
Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

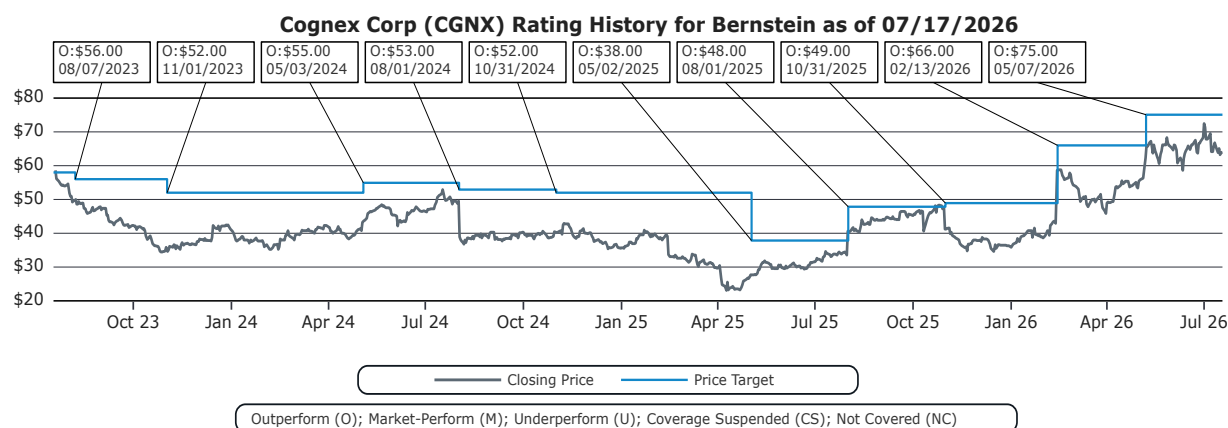
* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



Harmonic Drive Systems Inc (6324.JP) Rating History for Bernstein as of 07/17/2026**Keyence Corp (6861.JP) Rating History for Bernstein as of 07/17/2026****Shenzhen Inovance Technolo-A (300124.CH) Rating History for Bernstein as of 07/17/2026**



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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